JIAYIN, ZHANG

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EDUCATION

Nanjing University, China 09/2023 - Present

● Bachelor of Science in Atmospheric Science CGPA:4.56/5.0 Ranking: 3/78

- Core modules: Introduction to Atmospheric Science, Fluid Dynamics, Introduction to Environmental Science, Calculus I, Calculus II, Probability Theory and Mathematical Statistics, Ordinary Differential Equations, College Physics I, College Physics II, General Chemistry, Linear Algebra, Scientific Computing I

Hong Kong University of Science and Technology(Exchange Program) 09/2025- 12/2025

- Core modules: Atmospheric Chemistry(A), Practical Artificial Intelligence in Science(A), Air Quality Control and Management(A), Urban Air Pollution(A-)

RESEARCH INTERESTS

●Atmospheric chemistry; Machine learning; Environmental data science

PUBLICATIONS & PRESENTATIONS

- Aguilos, M., **Zhang, J.**, Belgado, M. L., Sun, G., McNulty, S., & King, J. (2025). Determining Early Warning Thresholds to Detect Tree Mortality Risk in a Southeastern U.S. Bottomland Hardwood Wetland. *Forests*, *16*(8), 1255. https://doi.org/10.3390/f16081255
- Characterization of Summer Hourly Extreme Precipitation and Attribution of Anthropogenic Discharge in the Middle and Lower Yangtze River Basin, **First Nanjing University Atmospheric Sciences Top Forum, Oral, Nanjing, China** 05/2025
- Linking Hydrological Stress to Ghost Forest Formation:Thresholds for Tree Mortality in Coastal Ecosystems, Maricar Aguilos, **Jiayin Zhang**, Miko Belgado, Ge Sun and John King, **AmeriFlux Annual Meeting**, Tuscon, USA 10/2025
- Modeling Evapotranspiration Dynamics Across Southeastern U.S. Ecosystems Using Flux Data and Machine Learning: M M Aguilos, G Sun, N Liu, **J Zhang**, M Belgado, S McNulty, J S King, **AGU Annual Meeting**, New Orleans, USA 12/2025

RESEARCH EXPERIENCE

- **Shanghai Urban Organic Aerosol Dynamics and Carbon Partitioning Analysis** 12/2025- present
  Advisors: Prof. Jianzhen Yu and Prof. Lixue Cheng, The Hong Kong University of Science and Technology
  Topic: Dynamic Regime Transitions and POC/SOC Partitioning of Urban Organic Aerosols in Shanghai
  - Analyzed long-term observational datasets of urban organic aerosols in Shanghai to characterize regime-dependent temporal dynamics and carbon partitioning behavior;
  - Applied PCA-GMM clustering and transition-matrix diagnostics to identify aerosol dynamic regimes, quantify residence time, stationary distribution, and switching behavior across episodes;
  - Evaluated inferred POC/SOC fractions through literature-based plausibility checks and external consistency analysis, with further plans for LSTM/PINN-based predictive modeling of regime evolution.
- **Sino-Finnish Helsinki Atmospheric and Earth System Scientific Expedition** 08/2025
  Advisor: Prof. Chao Yan, Nanjing-Helsinki Institute in Atmospheric and Earth System Science, Nanjing University

Topic: Investigation on Global Multi-site Particle Number Size Distribution

-Analyzed particle size distribution data from 16 long-term global stations (from ARM & EBAS database) using a nano-ranking approach to identify environmental variations;

- Conducted preliminary growth rate (GR), formation rate (J), and GDE analyses.

- **College Student Innovation and Entrepreneurship Program** 11/2024 - 07/2025

  Advisor: Prof. Anning Huang, School of Atmospheric Science, Nanjing University Provincial-level

  Topic: Characterization of Summer Hourly Extreme Precipitation and Attribution of Anthropogenic Discharge in the Middle and Lower Yangtze River Basin

  -Investigated spatiotemporal characteristics of extreme hourly precipitation events in the Yangtze River;

  -Conducted attribution analysis using CMIP6 model data to quantifying the impact of anthropogenic emissions.

- **GEARS Program** 01/2025 - 02/2025

  Advisor: Prof. Maricar Aguilos, Department of Forestry and Environmental Resources, North Carolina State University

  Title: Irreversible Impacts of Sea Level Rise and Prolonged Inundation on Carbon Fluxes and Tree Mortality: Drivers of Ghost Forest Formation

  - Applied Generalized Additive Models (GAMs), Mann–Kendall trend tests, and regression analyses in R to identify hydrological and carbon-flux drivers of tree mortality.

  - Used ROC analysis to establish early-warning thresholds and evaluate classification performance (AUC) for mortality risk prediction.

- **Early Research Training Program "Golden Seed" Project** 04/2024 - 10/2024

  Advisor: Prof. Yang Zhang, School of Atmospheric Science, Nanjing University

  Topic: Effects of Thermal Anomalies Over the Tibetan Plateau on the Southern Hemisphere

  -Conducted data visualization of 2-meter temperature, geopotential height, and wind fields using MATLAB;

  -Applied linear regression and random forest models to analyze the relationships among key climate variables;

  -Achieved first place and an "Excellent" rating in the final evaluation.

COMPETITIONS

- **2025 Asian Climate-SDG Technology Innovation HackathOn**

  - Finalist, invited to showcase the project at a roadshow in Hong Kong and went to Brazil to participate in the UN Climate Change conference (COP30).

  - Developed a novel urban digital twin platform designed to safeguard public health against climate-related risks like flooding and epidemics by integrating hydrodynamic, epidemiological, and community intelligence data.

EXTRACURRICULAR ACTIVITIES

- **Student Ambassador, Global Training Initiative of North Carolina State University** 02/2025 - Present

  -Managed communication between faculty and students, fostering global connections.

- **Standing Representative, Nanjing University Student Representative Congress** 10/2024 - Present

  -Advocated for student interests and contributed to initiatives aimed at improving campus environments.

- **Guzheng Performer, Nanjing University Traditional Chinese Music Ensemble** 10/2024 - Present

  -Performed as a Guzheng player, contributing to the preservation and promotion of traditional Chinese music.

- **Monitor, Nanjing-Helsinki Institute in Atmospheric and Earth System** 09/2024 - Present

  -Organized over ten activities, including class reunions, sports competitions, and intercollegiate bonding events.

AWARDS & HONORS

- **Renmin Scholarship in November 2025.**
- **Yan Baoling Overseas Exchange Scholarship Excellence Award** in June 2025.
- **Nanjing University Outstanding Student** in November 2024 and October 2025.
- **National Scholarship** in October 2024.
- **Outstanding Research Report in Nanjing University Summer Social Practice Program** in September 2024.
- **Nanjing University Freshman College Outstanding Student** in May 2024.

SKILLS

- **Languages:** Mandarin (native), English (Proficient): IELTS 7.5
- **Technical Skills:** Python, R, MATLAB, Office (PowerPoint, Word, Excel), LaTeX
- **Hobbies:** Volunteering (200+ hours), Badminton, Gym